



B.Tech. Degree IV Semester Examination April 2017

ME 15-1406 MANUFACTURING PROCESSES

(2015 Scheme)

Time: 3 Hours

Maximum Marks:60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What are the different allowances to be provided on pattern in sand casting process?
- (b) Which process would be preferred for mass production from among sand casting and die casting? Justify your answer.
- (c) What are the parameters which are generally tested and evaluated for sand selection in sand casting process? How are the tests done?
- (d) Differentiate between hot chamber and cold chamber die casting processes.
- (e) Explain the process of shell molding with figure.
- (f) With figures, explain the difference in location of burnish region and burr in blanking and punching components.
- (g) Explain the process of direct and indirect extrusion.
- (h) The press force needed for coining is much larger than embossing process for similar blanks. Justify the statement by explaining the process.
- (i) Differentiate between spot welding and seam welding.
- (j) Explain the process of Thermit welding.

PART B

(4 × 10 = 40)

- II. (a) What do you mean by gating ratio? What are the ideal gating ratios for pressurized and non-pressurized casting process? (2)
- (b) Explain the purpose of core prints and chaplets with figures. (4)
- (c) Explain Caine's method of riser design. (4)

OR

- III. (a) Explain the different methods to prevent impurities and turbulence in sand moulds. (5)
- (b) Explain all the elements of a proper gating system used in sand casting moulds. (5)

- IV. (a) Explain with figures the different types of centrifugal casting processes. (6)
- (b) What are the steps to be followed to minimize the cost of a cast product? (4)

OR

- V. Explain the various defects which can occur in sand casting process along with their remedies. (10)

- VI. (a) Differentiate between hot working and cold working operations. (5)
- (b) Define the term spring back in bending operation. Explain the methods used for eliminating spring back. (5)

OR

(P.T.O.)

- VII. (a) Explain with figures the different kinds of roll arrangements in rolling operation. (6)
- (b) Explain the mechanics of sheet metal cutting operation. (4)
- VIII. (a) What is the purpose of flux coating on welding electrodes? (3)
- (b) Compare the process of brazing and soldering. (4)
- (c) State the advantages and disadvantages of arc welding process using AC and DC current. (3)

OR

- IX. Explain the following welding processes. (4)
- (a) TIG welding. (4)
- (b) Projection welding. (3)
- (c) Laser beam welding. (3)